BLOCKCHAIN ENGINEERING

A Vishwateja

2303A510I5 - B16

Assignment Number: 01/12

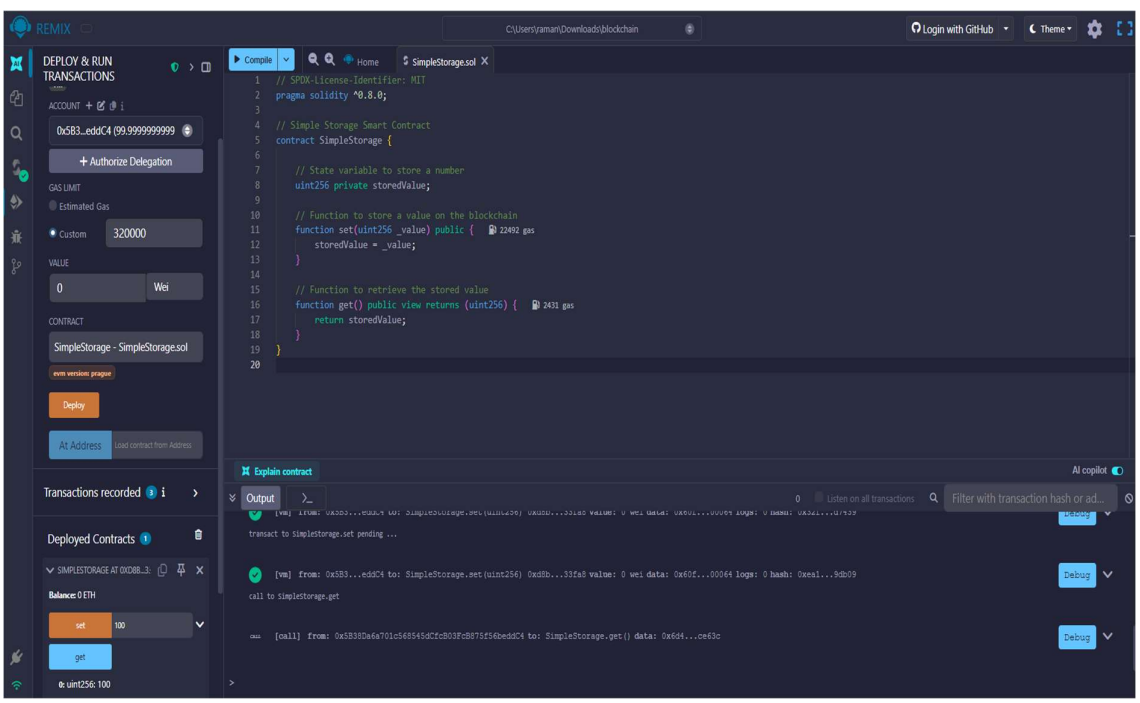
Q. No.	Question
1	**Title** Implementation of a Simple Storage Smart Contract using Remix IDE
	Objective To design, deploy, and interact with a **Simple Storage smart contract** using **Solidity** in the **Remix Ethereum IDE**, demonstrating basic blockchain concepts such as state variables, functions, and transactions.
	Tools / Software Required • Web Browser (Chrome / Firefox) • **Remix Ethereum IDE** (online) • Solidity Compiler (built-in with Remix) • JavaScript VM (Remix default environment)
	Theory (Brief) A **smart contract** is a self-executing program stored on the blockchain. The **Simple Storage contract** is a beginner-level Solidity contract that allows users to: • Store a value on the blockchain • Retrieve the stored value Remix IDE is a web-based development environment used to write, compile, deploy, and test Solidity smart contracts.

Requirements

- Create a Solidity smart contract named `SimpleStorage`
- Declare a state variable to store data
- Implement functions to:
  - Store a value
  - Retrieve the stored value
- Deploy the contract using Remix
- Test the contract with multiple inputs
- Add comments explaining the logic
- Capture screenshots of:
  - Contract compilation
  - Contract deployment
  - Function execution

Procedure / Steps

Step 1: Open Remix IDE

1. Open a web browser
2. Go to: https://remix.ethereum.org
3. Create a new file named:
4. `SimpleStorage.sol`

Code in Remix –